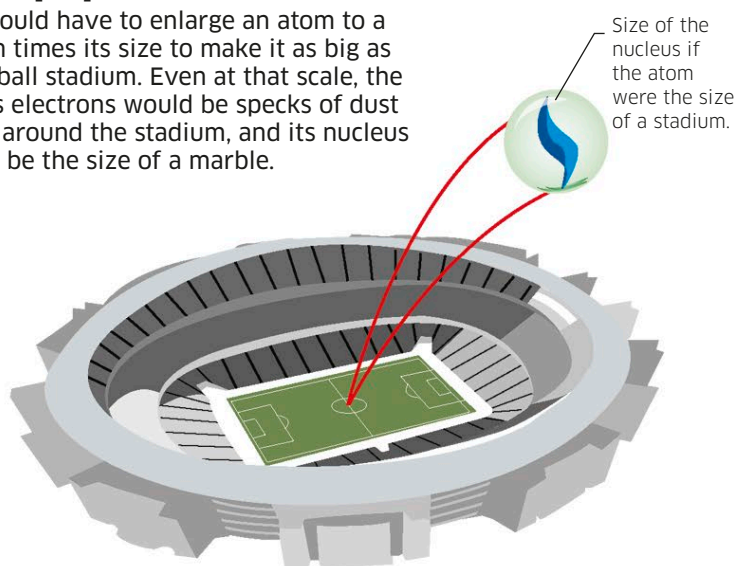


Atomic proportions

You would have to enlarge an atom to a trillion times its size to make it as big as a football stadium. Even at that scale, the atom's electrons would be specks of dust flying around the stadium, and its nucleus would be the size of a marble.



Atoms

You, and all the things around you, are made of tiny particles called atoms—particles so minuscule that even a small grain of sand is made up of trillions of them.

Atoms were once thought to be the smallest possible parts of matter, impossible to split into anything smaller. But they are actually made of even smaller particles called protons, neutrons, and electrons. Atoms join, or bond, in many different ways to make every different kind of material. A pure substance, consisting of only one type of atom, is called an element. Some familiar elements include gold, iron, carbon, neon, and oxygen. To find out more about the elements, see pp.28–41.

Atomic structure

The nucleus at the center of an atom is made of protons and neutrons. The protons carry a positive electric charge. The neutrons carry no charge—they are neutral. Around the nucleus are the electrons, which carry a negative electric charge. It is the force between the positively charged protons and the negatively charged electrons that holds an atom together.

Particles of an atom

Every atom of an element has the same number of electrons as it has protons, but the number of neutrons can be different. Below are the particles of one atom of the element carbon.

+		-
+		-
+		-
+		-
+		-
+		-
6	6	6
Protons	Neutrons	Electrons

The nucleus of a carbon atom has six protons and six neutrons.

The outer shell in a carbon atom holds four electrons.

Carbon atom

The number of protons in an atom's nucleus is called the atomic number. This defines what an element is like: each element has a different atomic number, as shown in the periodic table (see pp.28–29). For the element carbon, shown here, the atomic number is 6. An atom's number of electrons is also equal to its atomic number.

